

Year 9 Non- Calculator Exam Term 3 2011

Calculators are **NOT** to be used in this exam.

1. Convert 78% to a decimal.
2. Find 10% of \$35.40
3. If 20% of a number is 68, what is the number?
4. Find the GST paid on an item with a post GST price of \$121.
5. Decrease \$50 by 40%
6. It cost Diana \$60 to make a set of drawers. For how much would she need to sell them to make a profit of 15%?
7. John bought a shirt for \$20 and the next day the price increased to \$25. Find the percentage increase as a percentage of the original price.
8. What is the cost of 12 books at \$3.40 each.
9. If 800 sheep cost \$4000, how many can be bought with \$3000?
10. Increase \$36 by 25%.
11. Divide \$120 in the ratio 3 : 5.
12. Find the amount paid for an item if it had a marked price of \$17.50 before 10% GST is added to it.
13. Express \$66 as a percentages of \$200.
14. If the cost of 3 books of the same type is \$4.50, then what is the cost of 10 such books?
15. Harley bought a motorbike for \$25 000, then later sold it at a 20% loss. What was his selling price?

Section I	Section II	Section III	Total
/10	/10	/30	/50

SECTION I :**Literacy in Mathematics**

Fill in the blanks with the correct terms from the list words given below:

List Words

Wages, taxable income, superannuation, profit, salary, net pay, overtime, discount, commission, gross pay, budget, income tax, goods and services tax.

1. Tax paid to the government which is based on the level of income received. _____
2. Income usually calculated on the value of goods sold. _____
3. Amount after allowable deductions are subtracted from the gross pay. _____
4. A fixed amount paid for a year's employment. _____
5. The amount of pay before any deductions. _____
6. A plan for the use of income. _____
7. The amount of pay an employee receives after deductions such as income tax. _____
8. A tax of 10% added to the base price of most goods and services. _____
9. To reduce the price of goods sold. _____
10. Pay given to an employee often based on an agreed hourly rate. _____

SECTION II Multiple Choice : Circle the best response

1. Jan works for 2 hours at 'time and a half'. She earns \$36.00. Find her normal hourly rate.
 (A) \$12.00 (B) \$18.00 (C) \$24.00 (D) \$27.00

2.

Discount for bulk posting	
Express Delivery	5% off
Regular Delivery	7% off

The cost of posting one letter is 45 cents. A discount is given when many letters are posted at the same time. Neil wants to post 200 letters using Express Delivery. He receives the discount for bulk posting. How much will Neil pay?

- (A) \$4.50 (B) \$6.30 (C) \$83.70 (D) \$85.50

3. A compact-disk player which cost \$120 was later sold for \$90. The loss as a percentage of the cost price was

1. (A) 25% (B) 30% (C) $33\frac{1}{3}\%$ (D) 75%

4. The tax payable on taxable incomes in the range \$20 701 to \$38 000 is given by:

\$3060 plus 34c for each \$1 over \$20 700.

How much tax does Theresa pay if her taxable income is \$29 408?

- (A) \$2960.72 (B) \$6020.72 (C) \$9998.72 (D) \$13 058.72

5. John's normal rate of pay is \$7.50 per hour. He is paid the normal rate for the first 30 hours, time and a half for the next 5 hours, and double time after that. His wage for a 37-hour week is

- (A) \$277.50 (B) \$292.50 (C) \$303.75 (D) \$311.25

6. Dino works 7 nights per week in a coffee shop. Each night he works from 10:00pm until 2:30am. He earns \$8 per hour before midnight and \$12 per hour after midnight. How much does he earn in a week?

- (A) \$46 (B) \$252 (C) \$322 (D) \$378

7. What was the base price before 10% GST was added to give a retail price of \$26.95?

- (A) \$29.65 (B) \$24.50 (C) \$24.26 (D) \$26.68

8. 200 g of coffee sells for \$3.80. Which of the following is equivalent to this?

- (A) 1 kg for \$1.90 (B) 50 g for \$0.90
 (C) 300 g for \$5.70 (D) 500 g for \$9.05

9. Grant receives \$24 for delivering 400 pamphlets. He receives an additional \$12 per day for delivering in wet weather. One wet day he delivered 600 pamphlets. How much should he receive?

- (A) \$28 (B) \$36 (C) \$48 (D) \$54

10. When Angela bought a \$30 jacket at a discount store she only had to pay \$18. Calculate the percentage discount.

- (A) 12% (B) 40% (C) 60% (D) $60\frac{2}{3}\%$

SECTION III (Show all necessary working)

1. a) List an advantage of being paid a salary. 2

b) List a disadvantage of being paid a salary.

2. Daniel is paid \$15.40 an hour. What is his wage when he works 38 hours in one week? 2

3. Which is better value for money: 2
(Show working out)

a) 300g salmon at \$4.25 OR

b) 185g salmon at \$3.56

4. Tiffany was given a discount of $12\frac{1}{2}\%$ on a dress with a marked price of \$64. How much did she pay for it? 2

5. Rachel bought a painting for \$350 and later sold it for \$560. 2

a) How much profit did she make?

b) Find the profit as a percentage of cost price.

6. A salesman's wages are \$230 per week plus a commission of 4 % on his sales. How much will he earn in a week when he sells \$7000 worth of goods? 2

7. Scott receives a 12 % increase on his weekly wage of \$560. What will he now earn annually? (let 1 yr = 52 weeks) 3

8. Naomi bought a computer system on terms. The market price was \$2300. She paid a deposit of \$1200 and 12 monthly repayments of \$115.

a. How much did she owe after the deposit? 1

b. What was the total of the monthly repayments? 1

c. The extra she paid in (b) above was the interest charged on the amount owed. What was the percentage of interest charged? 2

9. Erin has a part-time job from 4 p.m. to 5.30 p.m. Monday to Friday. She also works from 9 am to 2 pm on Saturdays. Her normal weekday rate of pay is \$9.60 per hour and on Saturdays she is paid \$12.00 an hour.

a. Find her weekly gross pay. 3

b. If she works for 40 weeks in a year, what is her annual income? 1

10. From Tyson's gross annual income of \$51 634, the following deductions are made ; 5% in superannuation, \$196 Medical insurance per month and \$189 house payments per week. He pays \$16 856 in taxes for the year. Find:

a. how much superannuation does he pay each year. 1

b. how much medical insurance does he pay annually? 1

c. what is the total of his deductions? 2

d. what is his net yearly salary? 1

e. what is his net weekly salary? 2

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SECTION I :**Literacy in Mathematics**

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4. A fixed amount paid for a year's employment. salary
5. The amount of pay before any deductions. gross pay
6. A plan for the use of income. budget
7. The amount of pay an employee receives after deductions such as income tax. net pay
8. A tax of 10% added to the base price of most goods and services. goods and services tax
9. To reduce the price of goods sold. discount
10. Pay given to an employee often based on an agreed hourly rate. wages

SECTION II Multiple Choice : Circle the best response

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- (A) \$4.50 (B) \$6.30 (C) \$83.70 ☒ (D) \$85.50

3. A compact-disk player which cost \$120 was later sold for \$90. The loss as a percentage of the cost price was
☒ (A) 25% (B) 30% (C) $33\frac{1}{3}\%$ (D) 75%

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- (A) 12% ☒ (B) 40% (C) 60% (D) $60\frac{2}{3}\%$

SECTION III

(Show all necessary working)

1/30

1. a) List an advantage of being paid a salary. 2

any one of these {

- Permanent Employment
- Holiday and Sick Pay
- Superannuation
- Bonus as incentive
- Time off for working outside normal hours

- b) List a disadvantage of being paid a salary.

any one of these {

- During busy times additional hours worked without extra pay.
- very little flexibility in working hours (9am-5pm)

2. Daniel is paid \$15.40 an hour. What is his wage when he works 38 hours in one week? 2

$$\begin{aligned} & \$15.40 \times 38 \\ & = \$585.20 \end{aligned}$$

3. Which is better value for money: (Show working out) 2

- a) 300g salmon at \$4.25 OR

$$\frac{4.25}{300} = 1.42c/g$$

- b) 185g salmon at \$3.56

$$\frac{3.56}{185} = 1.92c/g$$

∴ (a) is better value

4. Tiffany was given a discount of 12½% on a dress with a marked price of \$64. How much did she pay for it? 2

$$\begin{aligned} & \$64 - (0.125 \times \$64) \\ & = \$56 \end{aligned}$$

$$\begin{aligned} & \$64 \times 0.875 \\ & = \$56 \end{aligned}$$

5. Rachel bought a painting for \$350 and later sold it for \$560. 2

- a) How much profit did she make?

$$\begin{aligned} & \$560 - \$350 \\ & = \$210 \end{aligned}$$

- b) Find the profit as a percentage of cost price.

$$\begin{aligned} & \frac{210}{350} \times 100\% \\ & = 60\% \end{aligned}$$

6. A salesman's wages are \$230 per week plus a commission of 4% on his sales. How much will he earn in a week when he sells \$7000 worth of goods? 2

$$\begin{aligned} & \$230 + (0.04 \times \$7000) \\ & = \$230 + \$280 \\ & = \$510 \end{aligned}$$

7. Scott receives a 12% increase on his weekly wage of \$560. What will he now earn annually? (let 1 yr = 52 weeks) 3

$$\begin{aligned} & \$560 + (0.12 \times \$560) \\ & = \$560 + \$67.20 \\ & = \$627.20 \end{aligned}$$

$$\therefore \text{new yearly wage} = \$627.20 \times 52$$

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$$= \$32614.40$$

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8. Naomi bought a computer system on terms. The market price was \$2300. She paid a deposit of \$1200 and 12 monthly repayments of \$115.

- a. How much did she owe after the deposit? 1

$$\begin{aligned} & \$2300 - \$1200 \\ & = \underline{\underline{\$1100}} \quad \checkmark \end{aligned}$$

- b. What was the total of the monthly repayments? 1

$$\begin{aligned} & \$115 \times 12 \\ & = \underline{\underline{\$1380}} \quad \checkmark \end{aligned}$$

- c. The extra she paid in (b) above was the interest charged on the amount owed. What was the percentage of interest charged? 2

$$\begin{aligned} \text{Interest} &= \$1380 - \$1100 \\ &= \underline{\underline{\$280}} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \therefore \text{Percentage Interest} &= \frac{280}{1100} \times 100\% \\ &= \underline{\underline{25.45\%}} \quad \checkmark \end{aligned}$$

9. Erin has a part-time job from 4 p.m. to 5.30 p.m. Monday to Friday. She also works from 9 am to 2 pm on Saturdays. Her normal weekday rate of pay is \$9.60 per hour and on Saturdays she is paid \$12.00 an hour.

- a. Find her weekly gross pay. 3

$$(\$9.60 \times 1.5 \times 5) + (\$12 \times 5)$$

$$= \underline{\underline{\$132}} \quad \checkmark$$

- b. If she works for 40 weeks in a year, what is her annual income? 1

$$\begin{aligned} & \$132 \times 40 \\ & = \underline{\underline{\$5280}} \quad \checkmark \end{aligned}$$

10. From Tyson's gross annual income of \$51 634, the following deductions are made; 5% in superannuation, \$196 Medical insurance per month and \$189 house payments per week. He pays \$16 856 in taxes for the year. Find:

- a. how much superannuation does he pay each year. 1

$$\begin{aligned} & \$51634 \times 0.05 \\ & = \underline{\underline{\$2581.70}} \quad \checkmark \end{aligned}$$

- b. how much medical insurance does he pay annually? 1

$$\begin{aligned} & \$196 \times 12 \\ & = \underline{\underline{\$2352}} \quad \checkmark \end{aligned}$$

- c. what is the total of his deductions? 2

$$\begin{aligned} & \$2581.70 + \$2352 \\ & + (\$189 \times 52) + \$16856 \\ & = \underline{\underline{\$31617.70}} \quad \checkmark \end{aligned}$$

- d. what is his net yearly salary? 1

$$\begin{aligned} & \$51634 - \$31617.70 \\ & = \underline{\underline{\$20016.30}} \quad \checkmark \end{aligned}$$

- e. what is his net weekly salary? 2

$$\begin{aligned} & \frac{\$20016.30}{52} \\ & = \underline{\underline{\$384.93}} \quad \checkmark \end{aligned}$$

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